

Circular Projection for Cyclic Compositional Data

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Transcriptomes often vary along periodic rather than linear coordinates. Standard displays can reveal such variation but do not link a sample’s position on a cycle to an ordering of the molecular features that define it. *clrcycle* applies a centered log-ratio transformation, learns a cyclic feature ordering by maximizing first-Fourier-mode variance, and projects samples onto the corresponding cosine and sine coordinates. Each sample receives a phase and a radius measuring the strength of its first-harmonic compositional signal. In the Hogenesch mouse circadian atlas, a supervised reference is contrasted with label-free selection by two-cycle reproducibility, which recovers phase in ten of twelve tissues and reveals tissue-specific changes in rhythmic program strength.

1. Introduction

Visualization is often the first step in interpreting high-dimensional transcriptomic data. Classical linear methods such as singular value decomposition and principal component analysis have long been used to summarize large expression matrices (Alter et al., 2000). More recent nonlinear methods, including diffusion maps (Coifman et al., 2005), UMAP (McInnes et al., 2018), and PHATE (Moon et al., 2019), have become standard tools for displaying cell states, developmental continua, and branching processes. In single-cell analysis, visualization is also closely tied to pseudotime inference: methods such as Monocle (Trapnell et al., 2014), reversed graph embedding (Qiu et al., 2017), and Slingshot (Street et al., 2018) infer ordered or branched trajectories and then use low-dimensional displays to show progression.

Many biological trajectories, however, are not fundamentally linear. Circadian time is periodic by definition. The cell cycle is a recurrent process. Some perturbation and regeneration experiments trace a return toward a starting state. These settings call for a display in which the primary latent coordinate is circular. Topological methods address this need by detecting loops and constructing circular coordinates from persistent cohomology (Silva et al., 2011). Such methods are attractive because they do not require a linear ordering and can represent genuinely cyclic sample geometry. They are also complementary to harmonic analyses of rhythmic transcription, where the dominant signal is often explicitly modeled by sine and cosine components (Hughes et al., 2009).

The method introduced here asks a different but related question. Instead of first finding a circular coordinate for samples, we ask whether the molecular features themselves can be ordered on a circle so that the dominant centered-log-ratio variation looks like a first Fourier mode. This gives a joint feature-sample visualization. The learned feature circle orders genes or modules by the phase at which their relative abundance

contributes to the cyclic signal. The sample projection then gives a phase-amplitude display: angle corresponds to phase, and radius corresponds to strength of the cyclic compositional contrast.

We call the approach *clrcycle*. The compositional step is important because many transcriptomic measurements are naturally interpreted through relative abundance after library normalization, and because compositional data are most naturally analyzed using log-ratio coordinates (Aitchison, 1982). *clrcycle* imposes a cyclic feature order and then uses the first circular Fourier mode as the visualization axis. The result is not intended to replace UMAP, diffusion maps, or topological loop detection. Rather, it is a targeted projection for cases in which a cyclic biological coordinate is already plausible and the goal is to obtain an interpretable sample phase plot together with a feature phase ordering.

This motivation is closely related to simplex-aware visualizations of gene specificity profiles. When each gene is summarized by weights over a finite set of biological states, the resulting vectors live on a simplex: raw Euclidean PCA can distort distances, while barycentric polygon plots can be interpretable but depend on the ordering of polygon vertices. A useful compromise is to move through log-ratio coordinates and then align an interpretable polygon to the observed geometry. The dependence of radial displays on circular coordinate or anchor order is a known visualization problem in star-glyph and RadViz-style plots, where the ordering of dimensions around a polygon can strongly affect the perceived shape and separation of displayed objects (Hu et al., 2021; Zhu et al., 2022). *clrcycle* adopts the same principle for cyclic transcriptomic structure: use log-ratio geometry first, then learn the polygonal feature order that makes the two-dimensional display biologically interpretable.

2. Results

2.1. A circadian benchmark in the Hogenesch mouse tissue atlas

We analyzed the mammalian circadian gene-expression atlas of Zhang et al. (2014). This study sampled 12 mouse tissues every two hours for two days, giving 24 samples per tissue. Because the sampling times are known, this dataset provides a direct benchmark: the *clrcycle* sample angle can be compared with circadian time, while the sample radius quantifies the strength of the first-harmonic compositional signal. We used the known times to construct a supervised reference and then assessed recovery from repeated measurements alone.

2.2. Supervised reference using time-associated probes

For a supervised reference, we selected 240 mapped probes with the strongest first-harmonic association to known circadian time in each tissue, applied the centered log-ratio transformation, learned a *clrcycle* feature ordering, and projected the samples onto the first Fourier mode of the ordered feature circle. The learned coordinates were aligned to circadian time only for reporting. Because the same time labels are used for feature selection and evaluation, this analysis is a descriptive upper reference rather than an independent validation.

In liver, the supervised projection agreed strongly with known circadian time ($R = 0.951$, mean absolute phase error = 1.03 hours; Figure 1). Across tissues, agreement ranged from $R = 0.943$ in kidney to $R = 0.983$ in skeletal muscle, with mean absolute phase errors from 0.59 to 1.16 hours.

The supervised projections exhibit a similar orientation of their elliptical sample traversals across tissues (Figure 2). This common tilt should not be interpreted as a biological finding: each projection is independently rotated and, if necessary, reflected to align its angle with known circadian time. The biologically informative differences are instead the shape of each traversal and its radius profile. Several peripheral tissues show a radius maximum near CT8–CT12 and a trough near CT16, whereas skeletal muscle has its largest radius near CT0 and a trough near CT16. These patterns summarize tissue-specific changes in the strength of the selected rhythmic compositional program over the circadian cycle.

The timing of these extrema is biologically plausible, but its interpretation requires care. In nocturnal mice, CT12 marks the transition into the subjective night and active phase. Thus, the broad CT8–CT12 radius maxima in several peripheral tissues are consistent with coordinated transcriptomic remodeling around that transition. The contrasting CT0 maximum in skeletal muscle is also credible: tissue-specific circadian phase programs, including markedly divergent profiles in liver and heart, have been observed in independent transcriptome studies (Storch et al., 2002; Zhang et al., 2014). Importantly, a radius maximum is not the peak time of a particular gene, pathway, or tissue clock. It is a rotation-invariant summary of the magnitude of the selected 240-probe compositional contrast, and therefore identifies when that supervised rhythmic program is most strongly expressed relative to its within-sample background.

2.3. Label-free recovery from repeated cycles

For each tissue, we selected 96 mapped probes whose expression varied reproducibly between the two sampled circadian cycles, without using circadian time labels. We then applied the centered log-ratio transformation, learned an *clrcycle* feature ordering, and projected the samples onto the first Fourier mode of the ordered feature circle. Circadian time was used only after fitting to align and evaluate the display; the arbitrary rotation and reflection of a two-dimensional circular projection do not otherwise affect the geometry.

In liver, the label-free projection traces the circadian cycle (Figure 3). Samples progress around the *clrcycle* plane in circadian order, and the feature circle arranges selected probes by inferred phase. The sample angle agrees strongly with known circadian time ($R = 0.942$, mean absolute phase error = 1.13 hours). The radius is not a confidence interval or fitted orbit; it is the magnitude of the first-harmonic *clr* contrast for each sample. Thus large radii indicate times at which the selected rhythmic log-ratio program is especially strong, while small radii mark relative troughs in that program.

The same label-free procedure was run across all 12 tissues (Figure 4). Ten tissues had phase agreement $R > 0.89$, ranging from $R = 0.904$ in aorta to $R = 0.958$ in brainstem, with mean absolute phase errors from 0.91 to 1.53 hours. Adrenal and brown fat were weaker ($R = 0.536$ and $R = 0.805$, respectively), illustrating that reproducible variation alone does not guarantee a dominant circadian program. The radius profile varied by tissue, suggesting tissue-specific “tides” in the strength of the first-harmonic circadian program.

3. Discussion

clrcycle resolves a cyclic transcriptomic program into two linked objects: a circular feature ordering and a phase-amplitude projection of samples. In the circadian atlas, sample angle recovers circadian phase, sample radius measures the strength of the rhythmic compositional contrast, and feature position summarizes the phase of the corresponding log-ratio program.

The two analyses separate a reference display from label-free recovery. Selecting probes by their association with circadian time yields uniformly high agreement, but uses the evaluation variable during feature selection. Selecting probes by reproducibility across the two sampled cycles removes this dependence and still recovers phase in ten tissues. The weaker adrenal and brown-fat projections show that reproducible variation is not necessarily dominated by a single circadian mode. This contrast makes the radius and trajectory shape interpretable as tissue-specific properties of the recovered program rather than consequences of a uniformly imposed feature set.

The compositional formulation is central to this interpretation. An arbitrary polygon can make features legible while scrambling their geometry, whereas PCA in log-ratio coordinates preserves geometry but does not yield phase-readable features. *clrcycle* searches for a feature order whose regular-polygon Fourier coordinates capture *clr* covariance, thereby

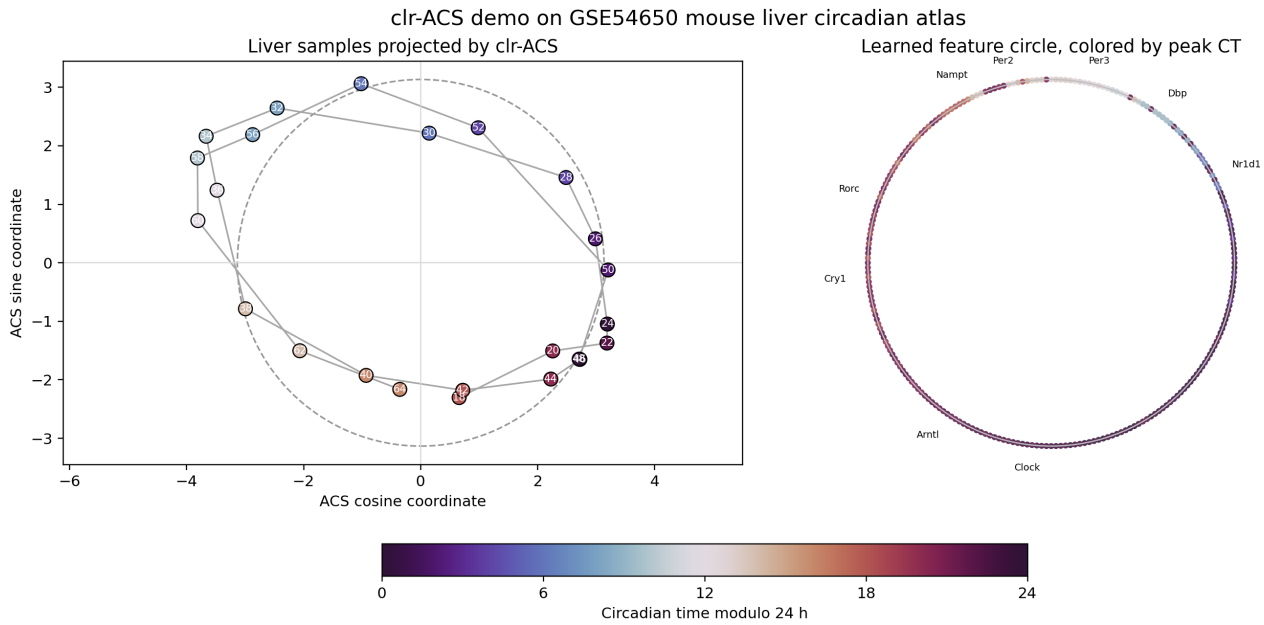


Figure 1: Supervised clrcycle projection of the GSE54650 mouse liver circadian time course. Probes were selected by their first-harmonic association with known circadian time. Left: liver samples projected onto the first circular Fourier mode after learning a cyclic feature order, colored by circadian time modulo 24 hours. The dashed circle is the median sample radius. Right: learned feature circle for the selected probes, with representative clock-associated genes labeled.

Table 1: Summary of circadian recovery across tissues. Peak and trough times refer to the largest and smallest clrcycle radii among the two-day samples, reported modulo 24 circadian hours.

Tissue	R	Error (h)	Peak CT	Trough CT
Adrenal	0.536	3.22	18	14
Aorta	0.904	1.53	12	16
Brainstem	0.958	0.91	12	6
Brown fat	0.805	2.04	20	2
Cerebellum	0.929	1.22	12	14
Heart	0.947	1.04	12	18
Hypothalamus	0.929	1.19	12	2
Kidney	0.949	1.06	10	16
Liver	0.942	1.13	10	16
Lung	0.943	1.14	10	16
Muscle	0.941	1.10	0	16
White fat	0.942	1.14	10	16

joining these two properties. The ordering is obtained by a discrete heuristic, but its objective is explicit and its projection is defined in log-ratio coordinates.

clrcycle complements nonlinear embeddings, pseudotime methods, and topological circular coordinates. Those approaches characterize neighborhoods, branches, or the existence of a loop in sample space. clrcycle instead tests a specific representation: whether an ordered set of compositional features supports a dominant first-harmonic projection. Its contribution is therefore not cyclic phase inference itself, but a feature-resolved phase-amplitude display for cyclic transcriptomic structure.

4. Methods

4.1. Input matrix and centered log-ratio transform

Let $X \in \mathbb{R}_{\geq 0}^{n \times D}$ be a sample-by-feature matrix after selecting a feature set. If any entry is nonpositive, we add $\alpha = 10^{-3} \min\{x_{ij} : x_{ij} > 0\}$ to every entry. Each sample is then transformed by the centered log-ratio map

$$z_{ij} = \log x_{ij} - \frac{1}{D} \sum_{k=1}^D \log x_{ik}.$$

The resulting matrix Z represents relative log-abundance within the selected feature set. We center its columns, writing $Z_c =$

clr-ACS sample projections across GSE54650 tissues (rhythmic feature selection)

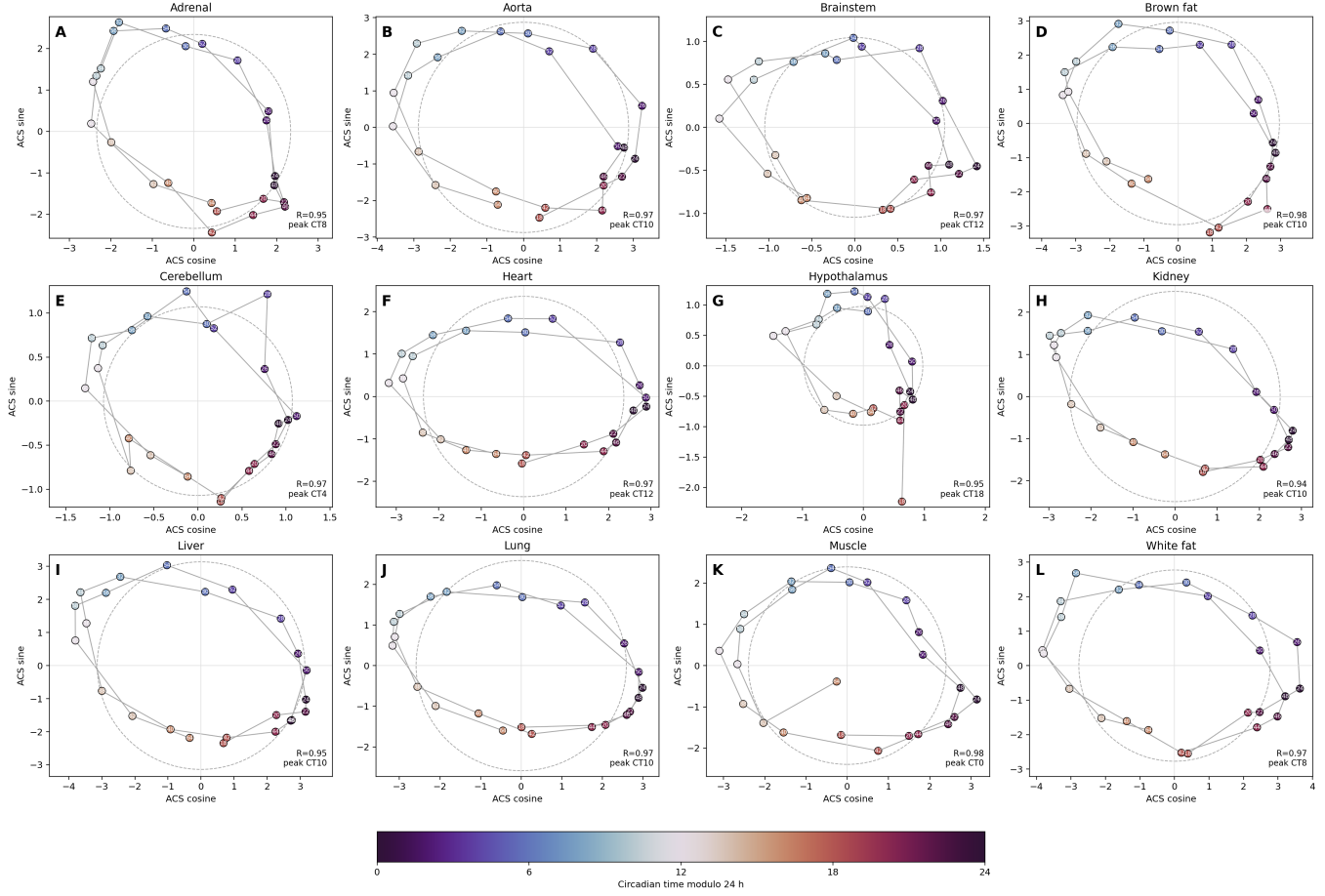


Figure 2: Supervised clrcycle sample projections across the Hogenesch mouse tissue atlas. Each panel uses probes selected by first-harmonic association with known circadian time. Samples are colored by circadian time, and black lines connect circadian-time centroids. The common direction of the colored traversal reflects the post-hoc alignment to time; tissue-specific trajectory shape and radius variation are the meaningful between-tissue features.

$Z - \mathbf{1}\bar{z}^\top$, before computing the feature covariance matrix

$$\Sigma = \frac{1}{n-1} Z_c^\top Z_c.$$

4.2. clrcycle feature ordering

For a candidate cyclic ordering π of the D features, place features at angles

$$\theta_j = \frac{2\pi j}{D}, \quad j = 0, \dots, D-1.$$

Define normalized first-harmonic vectors

$$c_j = \sqrt{\frac{2}{D}} \cos \theta_j, \quad s_j = \sqrt{\frac{2}{D}} \sin \theta_j.$$

For ordering π , let Σ_π be the covariance matrix with rows and columns permuted by π . The clrcycle objective is

$$F(\pi) = c^\top \Sigma_\pi c + s^\top \Sigma_\pi s.$$

The learned ordering is

$$\hat{\pi} = \arg \max_{\pi} F(\pi).$$

In the current implementation, this optimization is heuristic. If $\lambda_1 \geq \lambda_2$ are the two leading eigenvalues and e_1, e_2 their eigenvectors, we initialize the order by sorting features by the polar angle of $(\sqrt{\lambda_2}e_{2j}, \sqrt{\lambda_1}e_{1j})$. Starting from this order, we propose 4,000 uniformly sampled pairwise swaps and accept a swap only when it strictly increases $F(\pi)$. We use one start per tissue, with a fixed pseudorandom-number seed.

4.3. Sample projection

Given $\hat{\pi}$, each row z_i^c of Z_c is projected onto the first Fourier mode:

$$u_i = \sum_{j=1}^D z_{i,\hat{\pi}_j}^c c_j, \quad v_i = \sum_{j=1}^D z_{i,\hat{\pi}_j}^c s_j.$$

The polar representation is

$$\phi_i = \text{atan2}(v_i, u_i), \quad r_i = \sqrt{u_i^2 + v_i^2}.$$

The angle ϕ_i is interpreted as cyclic phase. The radius r_i is the strength of the first-harmonic compositional signal for sample i .

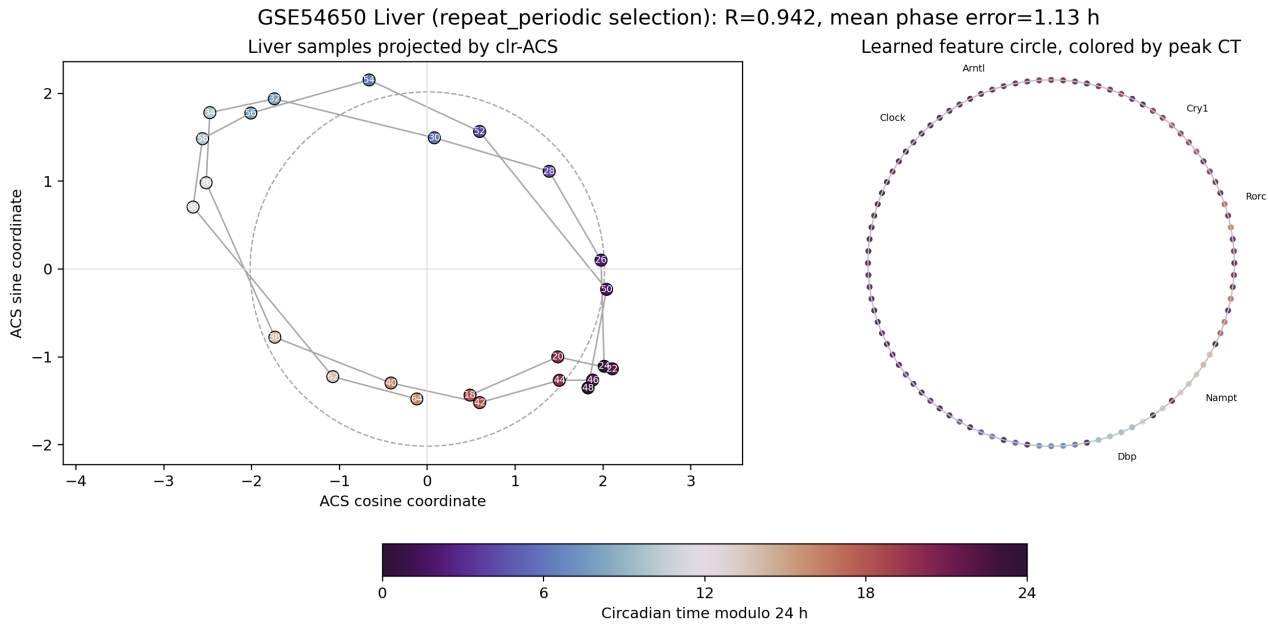


Figure 3: clrcycle projection of the GSE54650 mouse liver circadian time course after label-free feature selection. Left: liver samples projected onto the first circular Fourier mode after learning a cyclic feature order, colored by circadian time modulo 24 hours for post-hoc validation. The dashed circle is the median sample radius. Right: learned feature circle for probes selected by two-cycle repeatability, with representative clock-associated genes labeled.

4.4. Summary statistic

We report

$$\rho_{\text{clrcycle}} = \frac{F(\hat{\pi})}{\lambda_1 + \lambda_2},$$

where λ_1 and λ_2 are the two largest eigenvalues of Σ . This statistic compares the optimized first-harmonic variance with the maximum variance that any two-dimensional linear projection could capture. Values near one indicate that the dominant two-dimensional clr covariance is well represented by a first Fourier mode after cyclic feature ordering.

4.5. Phase validation

When known phases are available, write $\psi_i = \text{atan2}(v_i, u_i)$ and let ϕ_i^{ref} be the reference phase. For each of the two orientations $\psi_i^{(+)} = \psi_i$ and $\psi_i^{(-)} = -\psi_i$, we calculate

$$m_{\pm} = \frac{1}{n} \sum_{i=1}^n \exp\{i(\psi_i^{(\pm)} - \phi_i^{\text{ref}})\}, \quad \delta_{\pm} = \arg(m_{\pm}),$$

and plot aligned angles $\hat{\phi}_i = \psi_i^{(\pm)} - \delta_{\pm}$. We select the orientation with the largest circular agreement

$$R = |m_{\pm}|,$$

breaking an exact tie by smaller mean absolute circular error. Thus the rotation sets the plotted phase origin but does not change R ; only the optional reflection can change the agreement. We also report the mean absolute circular phase error, converted to hours for circadian data. This alignment is used only for validation and plotting; the clrcycle objective itself is unchanged by global rotation or reflection.

4.6. Hogenesch atlas analysis

GSE54650 expression data and GPL6246 probe annotations were downloaded from GEO. For the supervised reference analysis, we scored positively expressed, mapped probes by first-harmonic regression against known circadian time. Specifically, for $\vartheta_t = 2\pi(\text{CT}_t \bmod 24)/24$, we fit

$$\log x_{tg} - \overline{\log x_{\cdot g}} = \beta_{cg} \cos \vartheta_t + \beta_{sg} \sin \vartheta_t + \epsilon_{tg}.$$

Probes were ranked by the fitted fraction of sums of squares $R_g^2 = \|\hat{y}_{\cdot g}\|^2 / \|y_{\cdot g}\|^2$, with $\sqrt{\beta_{cg}^2 + \beta_{sg}^2}$ as the secondary criterion, and the top 240 were retained in each tissue. clr transformation, clrcycle ordering, and first Fourier projection then proceeded as described above. Circadian time was also used for post-hoc alignment, evaluation, and figure annotation. This analysis therefore quantifies the best-case display obtained when time-associated probes are supplied, rather than independent phase recovery.

4.7. Label-free repeated-cycle analysis

For each tissue, the 24 samples comprise two cycles with 12 matched sampling positions per cycle. For every positively expressed, mapped probe, we computed the log-expression repeatability score

$$q_g = \frac{\text{Var}_t \left[(y_{tg}^{(1)} + y_{tg}^{(2)})/2 \right]}{\frac{1}{4} \text{Mean}_t \left[(y_{tg}^{(1)} - y_{tg}^{(2)})^2 \right] + 10^{-8}},$$

where $y_{tg}^{(1)}$ and $y_{tg}^{(2)}$ are log expression in the two cycles at matched sampling position t . The top 96 probes by q_g ,

clr-ACS sample projections across GSE54650 tissues (repeat_periodic feature selection)

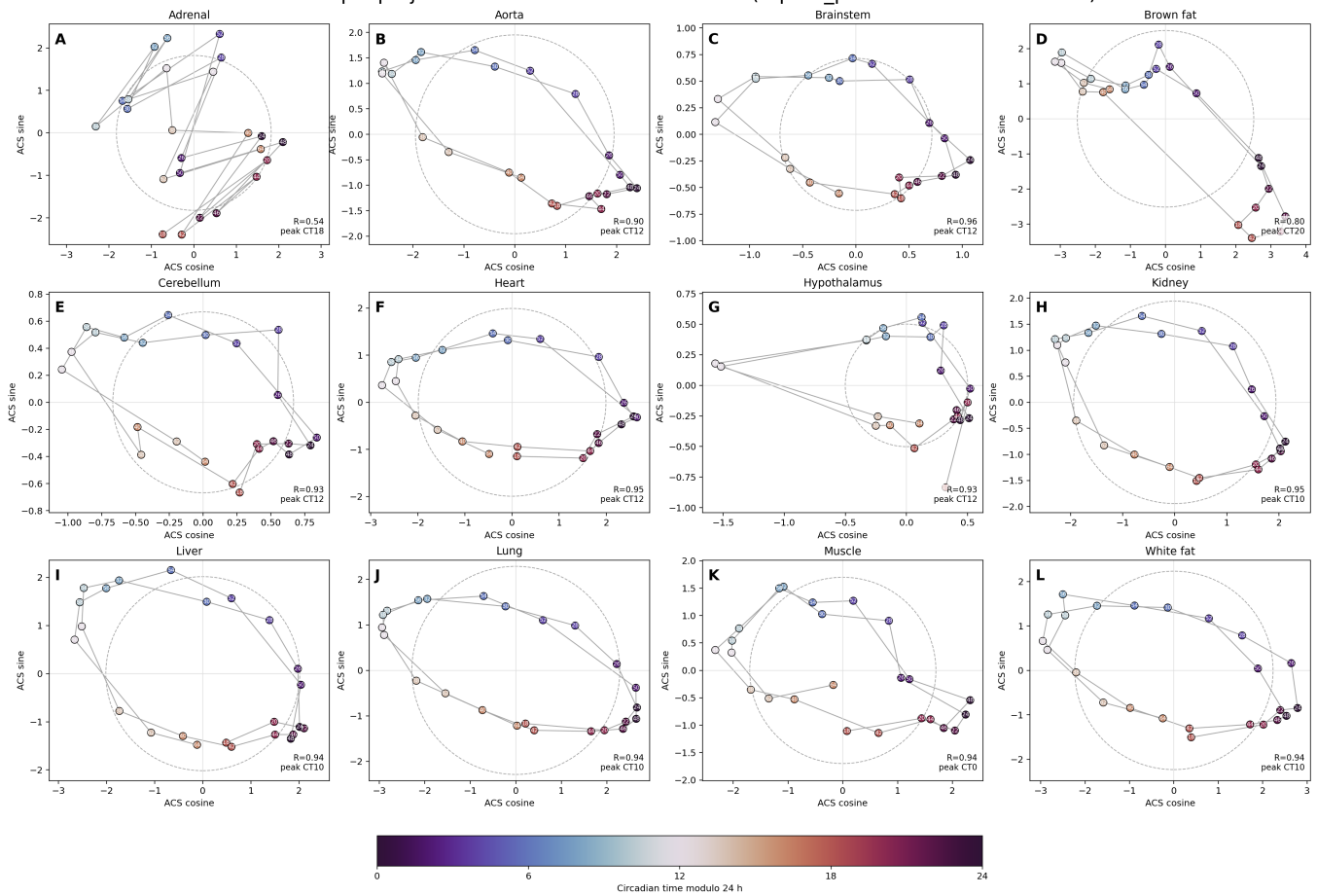


Figure 4: clrcycle sample projections across the Hogenesch mouse tissue atlas after label-free feature selection. Each panel shows one tissue from GSE54650. The 96 probes were selected using reproducibility across the two sampled cycles, not circadian time. Samples are colored by circadian time only for post-hoc validation, and black lines connect circadian-time centroids.

with between-cycle correlation as a secondary ranking criterion, were retained. This fixed panel size regularizes the feature set uniformly across tissues. Thus feature selection uses repeated-measurement structure but no circadian-time values. Each tissue was then analyzed separately by clr transformation, clrcycle ordering, and first Fourier projection. Circadian time was used only for post-hoc alignment, evaluation, and figure annotation. The all-tissue figure and summary table were generated from results/all_tissues_unsupervised_repeat_periodic_96/.

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